

- 3 A binary star system shown in Fig. 3.1 consists of two stars orbiting around their common centre of mass, P. One such binary star system is known as Sirius where the mass of Sirius A is twice the mass of Sirius B. The mass of Sirius B is 1.99×10^{30} kg.

Sirius A is at a distance of $\frac{1}{3}D$ from P. Sirius B is at a distance of $\frac{2}{3}D$ from P.

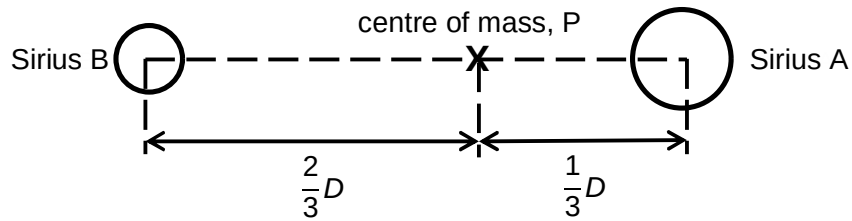


Fig. 3.1

- (a) State *Newton's law of gravitation*.

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- (b) The orbital period of this binary star system is 50 years. Determine the distance D between the binary stars, assuming that the stars move in circular orbits.

distance $D =$ m [4]

- (c) Determine the speed of a probe, originally at rest at infinity, when it reaches the centre of mass of the star system.

speed = m s^{-1} [3]